



An independent and self-motive undergraduate engineer seeking a full-time position in the field of **Electronics, Software, Mechanical, Automation, IoT, Robotics**, Where I can apply knowledge and skills for continuous improvements.

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Skill Highlights

- Skill Highlights
- Project management
- Strong decision maker
- Complex problem solver
- Creative spirit
- Innovative
- Service-focused
- Fast learner

Languages

Malayalam – C2

English – C1

Hindi – A1

Experience

August 2022 - Present

Field Application & Integration Engineer - ZENTRON LABS PVT, BANGALORE, India

- Examined the TMCL software to test the stepper motor driver's functionality.
- Perform hardware validation and testing using oscilloscopes, logic analyzers, and other diagnostic tools to ensure system functionality and reliability.
- Developed detailed system test plans and integration of Machine.
- Testing of system controller, Limit sensors, TMC Motor Driver, Rotary and linear sensors for maintain the quality of machine built.
- Provided technical support to customers for product selection, installation and troubleshooting.
- Created detailed reports and RCA on Vision system performance in the field.
- Perform root cause investigation and debugging of failed or returned Machines.
- Lead a team of junior engineers and provide technical guidance and mentor ship.
- Identified areas for improvement in existing products based on customer feedback.
- Conduct performance testing and validation of embedded system to meet industry standards and regulatory requirement.
- Support Customers with testing and technical issues to solve problems and configure product designs.

Programming Languages

- C++
- Python
- HTML
- Arduino (C)
- Angular.js Beginner Level
- Node.js Beginner Level
- PHP
- Java script, Java Beginner Level
- Ladder logic
- MATLAB
- SQL
- CSS

Professional Skills

- Embedded & Electronics Projects
- Marketing
- Web development
- Project management
- Photo shop
- Ki-Cad
- Frizzing (Simulation and circuit design tool)
- Solid works
- Premier Pro
- Robotics & Automation
- Embedded programming (Samc32, ATMEL)

August 2021 – Jan 2022

Mechatronics Engineer – Luxin Autonomy, Kochi

- Exposure to electric power train components such as motor, HV battery, DC-DC converter and charger.
- Regenerative braking section.
- Plan, define, and execute autonomous vehicle project works with raspberry pi using Open CV and Tenser-flow lite with no supervision.
- Test and supervise electrical and electronic components for EV.
- Experience working with drafting software such as Solid works.
- Mentored junior interns from different universities.
- Identify and repair faults on wiring harness and electrical systems.

Internship

Sept 2018

ROBOTICS INTERN - Sristhi Robotics, Kochi

- Arduino advanced projects.
- Raspberry-pi python projects.
- IOT (internet of things, Esp32, Node MCU)

Apr 2020

ELECTRICAL INTERN – VOLTECH A-Grade, Kochi

- Based on Power electrical wiring and devices.

Education

B Tech in Mechatronics Engineering – MGM College of Engineering and Pharmaceutical Sciences, Kerala, Malappuram.

- CGPA 7.48

Higher Secondary (Computer Science) – Cardinal HSS, Thrikkakara.

- 84.3 %

Certificates

ROBOTHON IEEE ROBOTICS & AUTOMATION, KOLLAM (National Level Robotics Hackathon).

CRASH COURSE ON PYTHON (GOOGLE).

TWICE STATE LEVEL WINNER FOR ELECTRONICS COMPETITION (Embedded system and circuit making competitor).

VOLTECH A-GRADE (ELECTRICAL CONTRACTORS).

EMBEDDED SYSTEM WORKSHOP (Arduino, raspberry pi, IoT).

Personal Projects

- **Application of python programming**
 - Raspberry Pi Based Autonomous Vehicle using OpenCV/ TensorFlow.
 - Automatic Fire Extinguisher Using Open-CV Image Processing.
 - Serial Python Jarvis projects (Voice assistance).
- **Drone (UAV's) Made from scratch** - Calibrated the drone equipment, did regular maintenance checks to maintain peak performance, and troubleshoot any technical issues that might have arisen.
- **Home automation and sensor data monitoring using Web server domain** - Used SQL, JavaScript, HTML for the Data monitoring.
- **IoT based Projects** – The popular development platforms like Arduino, Raspberry Pi, ESP 8266, ESP 32/CAM etc were used for the Projects.
- **MQTT Projects** - Adafruit io based Smart home system using Arduino WiFi.
- **Electronics Projects** - These projects belong to a wide range of categories such as Embedded, Power Electronics, Analog, Digital Electronics, Audio and Internet of Things.

Freelancing Projects

- **Web Page Designing** - Developed and hosted in crazydomains.in for an Australian customer by using HTML, PHP, JAVA, CSS etc.
- **Projects based on embedded systems (ATMEGA 328, ESP-32, ESP-8266 etc)** – Vehicle speed detection alert and IoT image capturing, Image processing using python Open-CV, Online web home automation, Sensor data monitoring using web domain and SQL, Li-Fi communication using shift register etc.
- **Mat-lab Simulation**
 - IPv4 UDP Checksum (Input will be DATA and output of the code will be 16 bit binary number).
 - Examined the time required for Image transmission of JPEG image by using different quality factor.
 - Simulation of Bernoulli Loss model with Packet loss.
- **Circuit Designing** – Single and double side PCB Layout development for freelancing Embedded Projects using Ki-Cad software.
- **Engineering Project report** - Created detailed report based on Mat-Lab projects and embedded projects for foreign universities.
- **Photoshop & Premier Pro** – Establish and manage a freelance graphic design, which includes Logos, Videos, and Templates.
- **Madrobotica (Not Registered)** - To aid in technical research and development, MadRobotica creates self-learning kits for researchers, students, corporations, institutes, and hobbyists.

Academic Projects

- **Disaster management bot with firefighting technology – (Final Year Project)**

Raspberry Pi with the OpenCV have been suggested as a fire detection approach for both day and night. The water sprayer is automatically aimed at the fire using computer vision technology. IoT based perfect sensor data monitoring.

Link- <https://youtu.be/DPOpSHq8wgl?si=T6ww2rAfmVSH3qLx>

- **IoT Based Automation And Monitoring – (Design Project)**

The robot module's functions include looking for living people, gathering data from the air quality control module, localizing it in relation to fixed nodes, and sending that data. The squad can view the data in real time via a website.

Link- https://drive.google.com/file/d/1GTA8z9dqRM-iii1bXp4DqPengelJW_X/view?usp=drive_link